

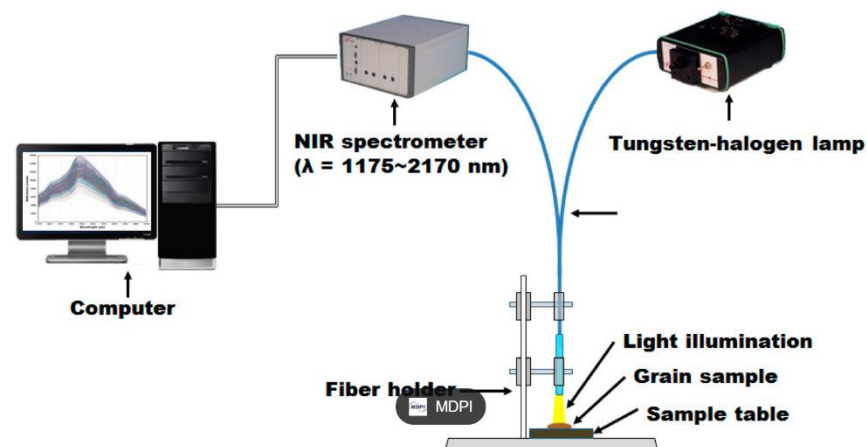
CHAPTER 11

Optical Properties of Phonons

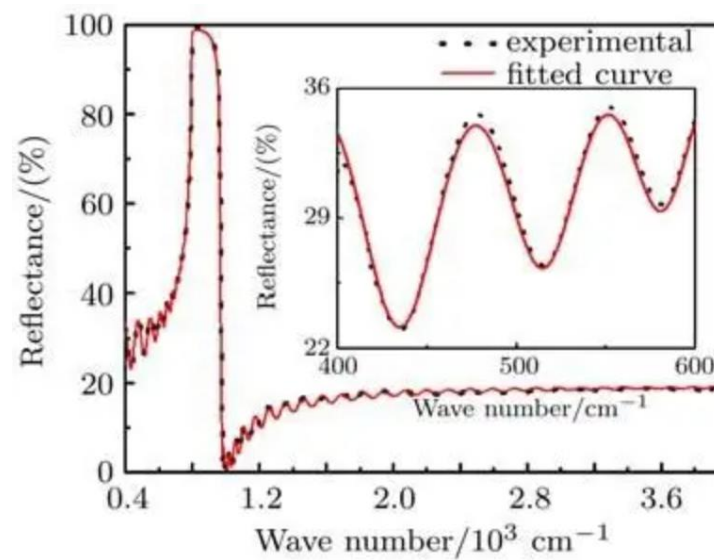
Contents

- In this chapter we will discuss the optical properties of phonons.
- We start with the properties of bulk materials, then the properties of materials with reduced dimensionality.

Infrared reflection spectrometer



$$R(\omega) = \frac{I_r}{I_i} = \frac{[n(\omega) - 1]^2 + \kappa^2(\omega)}{[n(\omega) + 1]^2 + \kappa^2(\omega)}$$



11.1 Phonons in Bulk Semiconductors

11.1.1 Reflection Spectra

- The reflectivity reaches values above 0.9 at room temperature and for the polarization $\mathbf{E} \parallel \mathbf{c}$ where the Γ_1 phonons are dipole allowed.
- The spectra in Fig. 11.1b to d have been deduced from the reflection spectrum with the help of the Kramers–Kronig relations and other formulae.
- The transverse and longitudinal mode frequencies coincide very well with the maxima of $\varepsilon_2(\omega)$ and loss function $\text{Im} \{-1/\varepsilon(\omega)\}$, respectively.
- It should be noted that in samples with (partly) ionic binding, optical phonon modes may exist, which do not carry a dipole moment and therefore do not couple to the radiation field, i.e. they are not optically active.

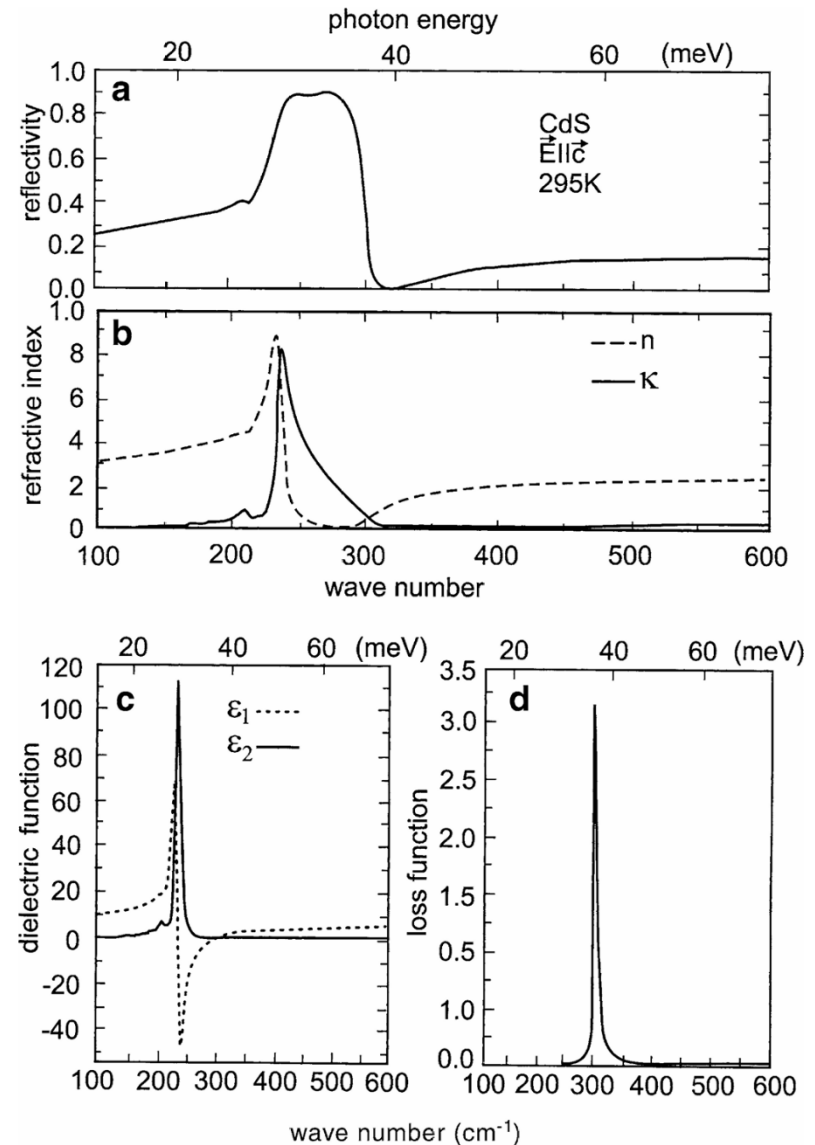
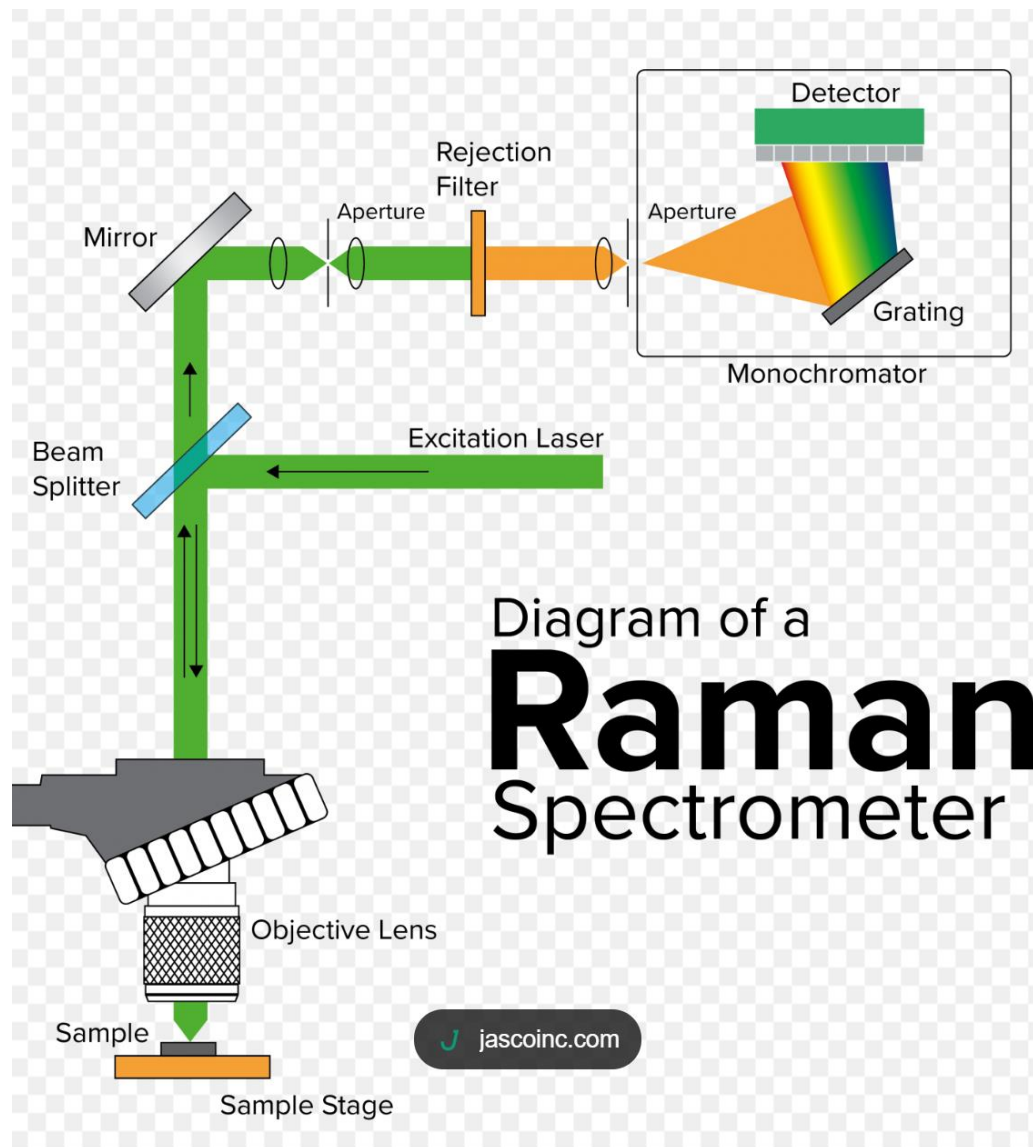


Fig.11.1a the reflection spectrum of CdS in the IR around the phonon resonance.

- Raman spectroscopy is a form of vibrational spectroscopy that provides detailed chemical and structural characterisation. The Raman technique has the benefit of being non-destructive and requiring no sample preparation
- Chandrasekhara Venkata Raman first observed Raman scattering in 1928 and subsequently received the 1930 Nobel Prize in Physics for his discovery
- Raman spectroscopy can be used to analyse the make-up of materials
- In Raman spectroscopy, focus a single colour of light onto a sample, measure the way that light interacts with the material to gain information about it. This gives us an understanding of its chemistry and structure
- We graphically display the results of Raman spectroscopy measurements as Raman spectra. The y-axis represents the intensity of the scattered light, while the x-axis represents the energy (frequency) of light. We are interested in the shift in frequency of the Raman-scattered light, so we plot the x-axis frequencies relative to that of the laser. We label the x-axis as the Raman shift (shown by the units cm^{-1}).



What information can I get from Raman spectroscopy



Characteristic frequencies



Identify the material using the spectral fingerprint



Variation in band position



Compressive or tensile stress/strain



Variation in intensity



Concentration or amount of material



Variation in band width



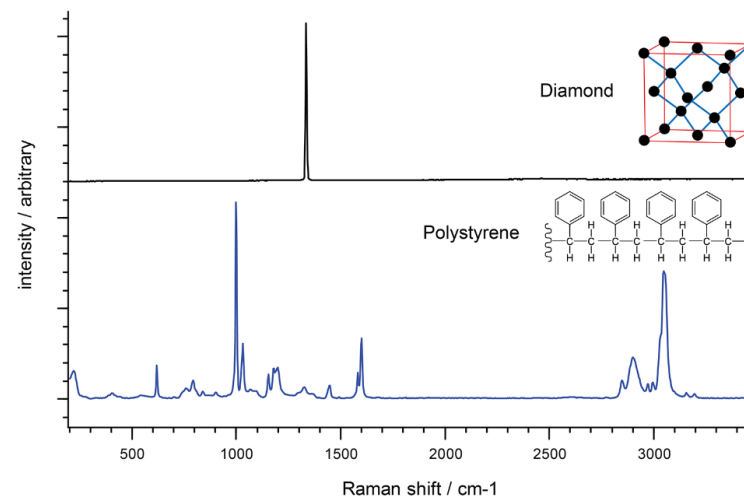
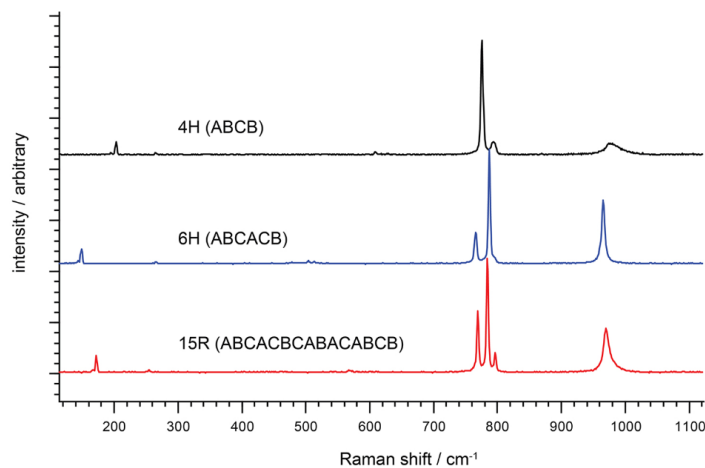
Uniformity of bond lengths e.g. crystallinity



Variation in intensity with changing laser polarisation



Crystallographic orientation



Raman spectra of diamond and polystyrene. The Raman spectrum of polystyrene is more complex than that of diamond due to differing bond types.

■ 11.1.2 Raman Scattering

Raman scattering with phonons is the inelastic scattering.

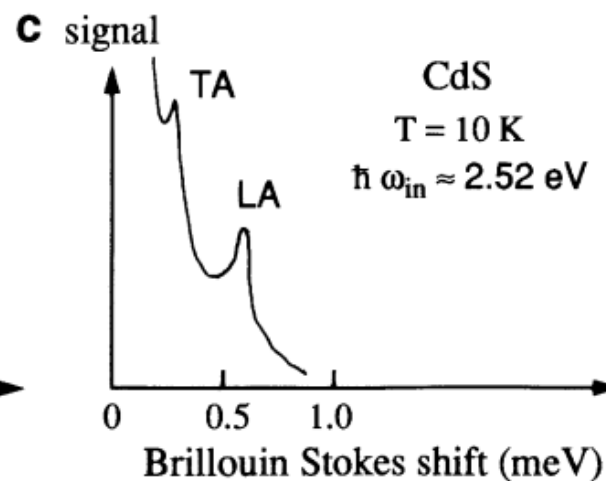
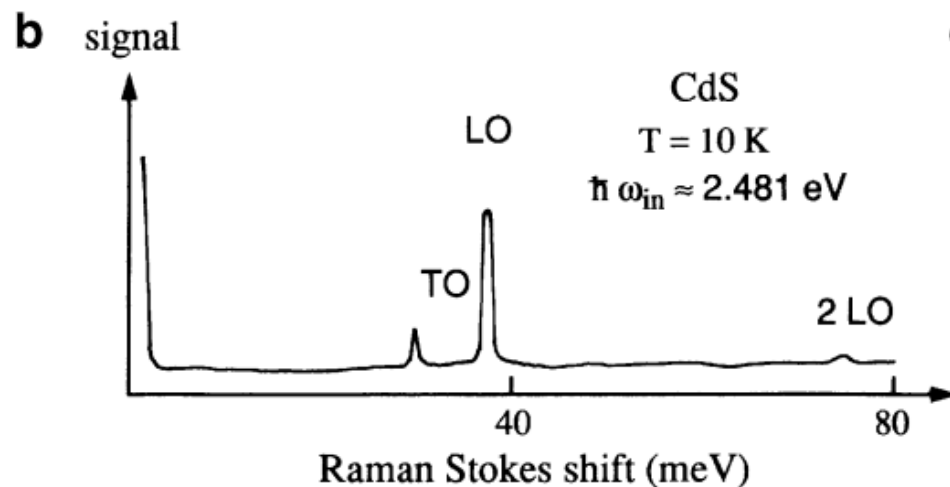
➤ Energy and (quasi-)momentum conserved:

$$\hbar\omega_R = \hbar\omega_i \pm \hbar\Omega_{ph}, \mathbf{k}_R = \mathbf{k}_i \pm \mathbf{k}_{ph}$$

➤ The selection rules are given in a classical description by the Raman tensor and in quantum mechanics by the transition matrix elements.

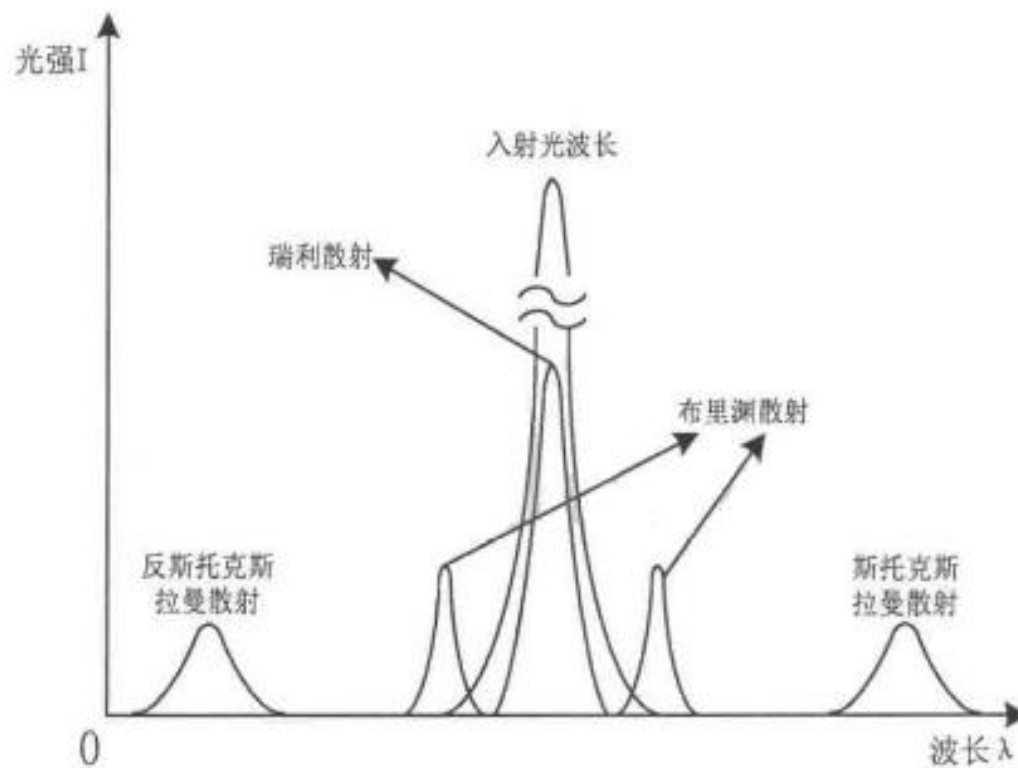
➤ The intensity ratio of Stokes to anti-Stokes emission:

$$\frac{I_s}{I_a} = \left(\frac{\omega_i - \Omega_{ph}}{\omega_i + \Omega_{ph}} \right)^4 \cdot \frac{N_{ph} + 1}{N_{ph}} = \left(\frac{\omega_i - \Omega_{ph}}{\omega_i + \Omega_{ph}} \right)^4 \cdot \exp \left\{ \frac{\hbar\Omega_{ph}}{k_B T} \right\}$$



■ 11.1.4 Brillouin Scattering

➤ Brillouin scattering is the analog of Raman scattering for acoustic phonons



■ 11.1.5 Surface Phonon Polaritons

- Surface polariton modes exist in the range between the transverse and longitudinal eigenmodes.
- These modes can be observed by attenuated total reflection (ATR) .
- Fig.11.5 shows the ATR spectra and in b the dispersion relation of the surface phonon polariton in GaP .

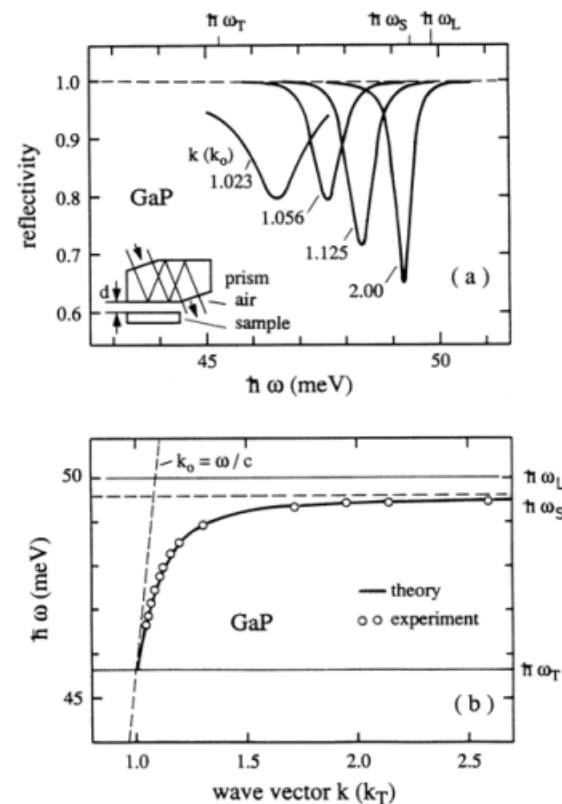


Fig. 11.5. Spectra of attenuated total reflection (a) in the region of the optical phonon resonances of GaP and the resulting dispersion relation of the surface phonon polariton (b). Compare with Fig. 11.4 for the bulk modes. According to [72M1]

11.1.6 Phonons in Alloys

- Alloys like $\text{Ga}_{1-y}\text{Al}_y\text{As}$, $\text{CdS}_{1-x}\text{Se}_x$ or $\text{Zn}_{1-y}\text{Mg}_y\text{S}_{1-x}\text{Se}_x$.
- We concentrate here on ternary alloys of two binary compounds.
- How the phonon modes behave in an alloy:
 - Amalgamation type: the two equivalent branches overlap energetically.
 - Persistent mode type: the two equivalent branches do not overlap energetically.
 - The acoustic phonons are always of the amalgamation type.
 - For optical phonons both cases can appear.
- Isotope mixtures present some (weak) alloying or disorder, contributing to a broadening of phonon resonances.

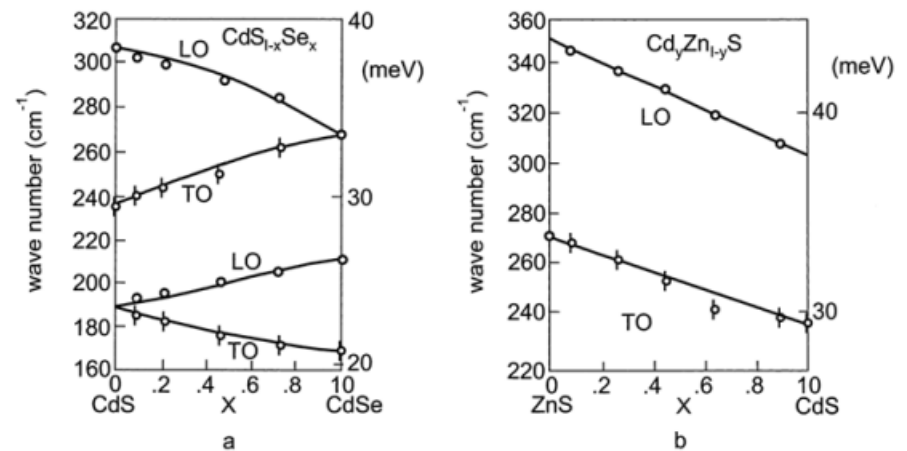


Fig. 11.6. The composition dependence of the optical phonon frequencies for a persistent mode, type alloy, here $\text{CdS}_{1-x}\text{Se}_x$ (a) and an amalgamation, type alloy, here $\text{Zn}_{1-y}\text{Cd}_y\text{S}$ (b) [68C1]

■ 11.1.7 Defects and Localized Phonon Modes

- The absorption spectrum of a GaAs: Si in the energy range above the optical bulk phonon modes is shown in Fig. 11.7
- Most of the peaks can be identified as local phonon modes of the centers or complexes indicated by arrows.
- Localized phonon modes can also be observed in Raman scattering or in luminescence or absorption in the visible as side-bands.

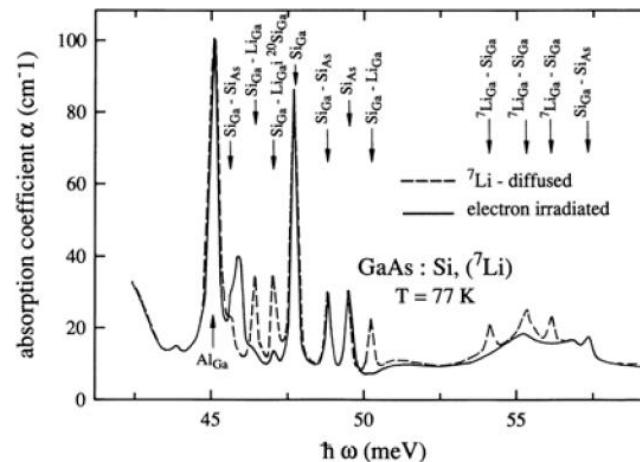


Fig. 11.7. Absorption spectrum of a GaAs: Si, Li, doped sample, showing localized phonon modes [71S1]

11.2 Phonons in Superlattices

Heterostructures like superlattices including: AlAs/GaAs or CdS/CdSe

■ 11.2.1 Backfolded Acoustic Phonons

- The rules for backfolding or confinement of phonons in a superlattice are essentially the same as in alloys.
- Backfolding always applies for acoustic phonons .

Fig.11.8: Comparison between the experimental and calculated dispersion relations.

- Under favorable conditions it is even possible to observe backfolded acoustic phonons directly in the IR.

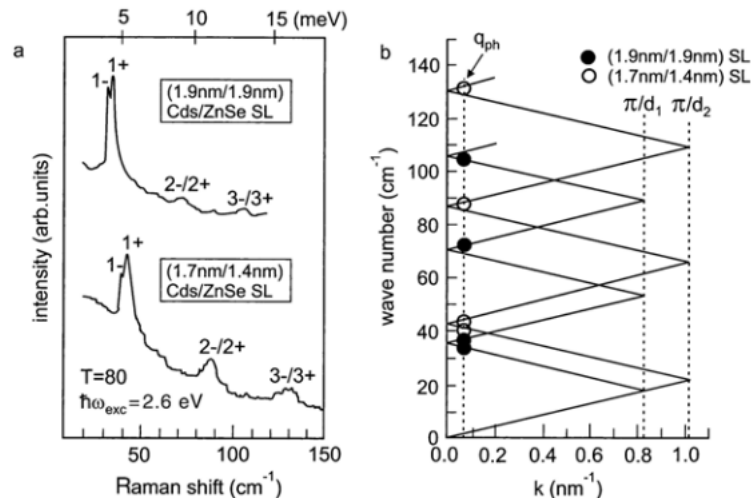


Fig. 11.8. Raman spectra of the first three backfolded longitudinal acoustic phonons (FLAPS) in two different CdS/ZnSe superlattices (a) and their calculated dispersion relation together with the data points from (a) (b) [01D1]

11.2.2 Confined Optic Phonons

- The CdS/ZnSe system is a persistent mode type.
- There are confined optical phonon modes in the two different sublattices.
- Fig.11.9a : the amplitudes decay extremely fast beyond the boundary.
- Fig. 11.10: By measuring the eigenfrequencies of confined phonons of different order and in superlattice layers of different thicknesses.

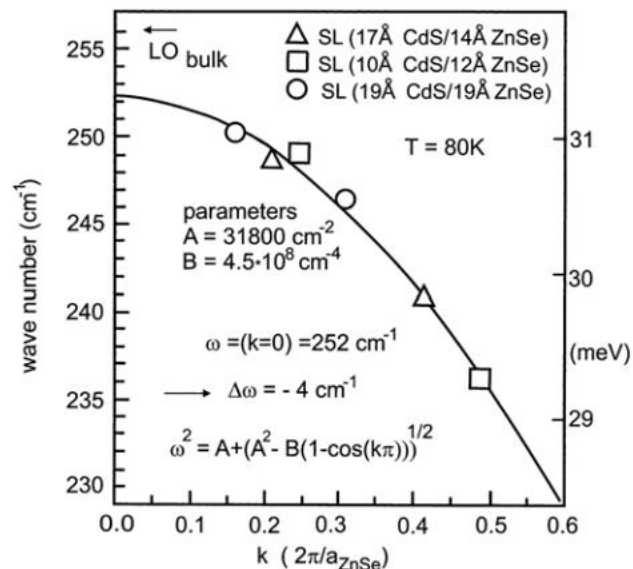


Fig. 11.10. The dispersion relation of LO phonons of ZnSe over one half of the first Brillouin zone deduced from confined optic phonons in a CdS/ZnSe superlattice [01D1].

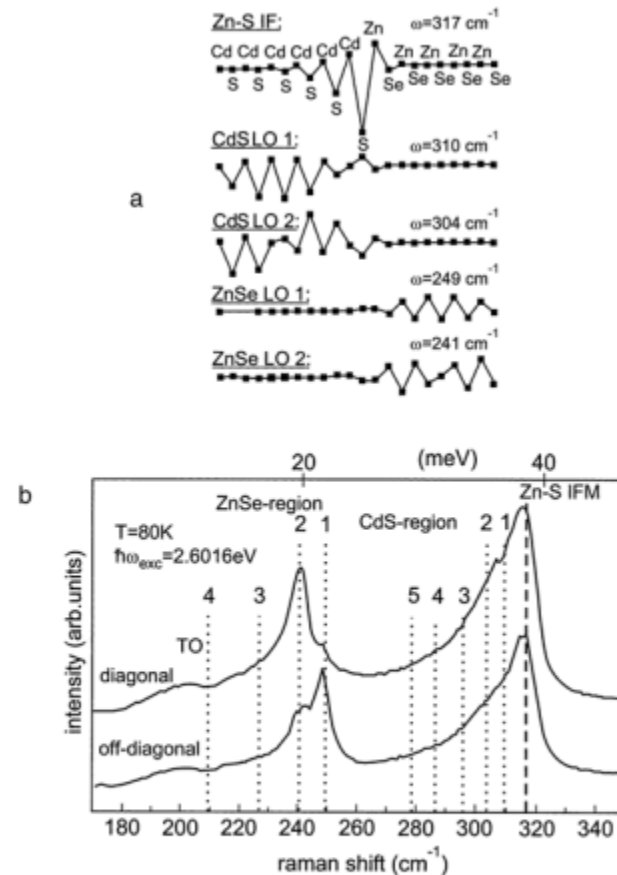


Fig. 11.9. Calculated atomic displacement patterns for the first two confined longitudinal optical phonon modes in a (CdS)₆/(ZnSe)₅ superlattice and of the ZnS interface mode (a) and the Raman spectra for parallel (diagonal) and orthogonal (off-diagonal) polarization with the calculated frequencies indicated by dashed vertical lines (b) [01D1]

■ 11.2.3 Interface Phonons

- Interface phonons may appear in some quaternary superlattices like CdS/ZnSe or BeTe/ZnSe.
- Phonon modes occur at the interface between two materials with amplitudes decaying rapidly on both sides.
- In the CdS/ZnSe system, two bonds can appear at the interface .
- CdSe interface phonon modes are difficult to detect .
- The ZnS mode involves the light cation and anion masses and different from CdSe.

■ 11.3 Phonons in Quantum Dots

- In quantum dots, phonons may be confined in all three directions of space similarly to charge carriers or excitons.
- The acoustic phonons show a discrete energy spectrum starting at finite energy.
- The appearance of the various modes is obviously closely linked to the boundary conditions of the dot.
- The confined acoustic phonons shift in energy according to R^{-1}